

Introduction to Computer Organisation and Architecture UCCD1133



Chapter 3 Basic Concept of Logic

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Outline

- SOP, POS & standard form SOP, POS
- Minterm & maxterm notation
- Combination circuit classification and analysis

Standard Form of Boolean Expression

- All Boolean expression can be converted into two standard form:
 - The sum-of-product form (SOP)
 - The product-of-sum form (POS)
- These standardization makes evaluation, simplification and implementation of Boolean expression more systematic and easier

Standard (Canonical) Forms of Logic Expression

- SOP (Sum of Product) is defined as two or more product terms are summed by Boolean addition expression. Example :

✓ $AB + ABC$

Sum of product

Product term/ minterm

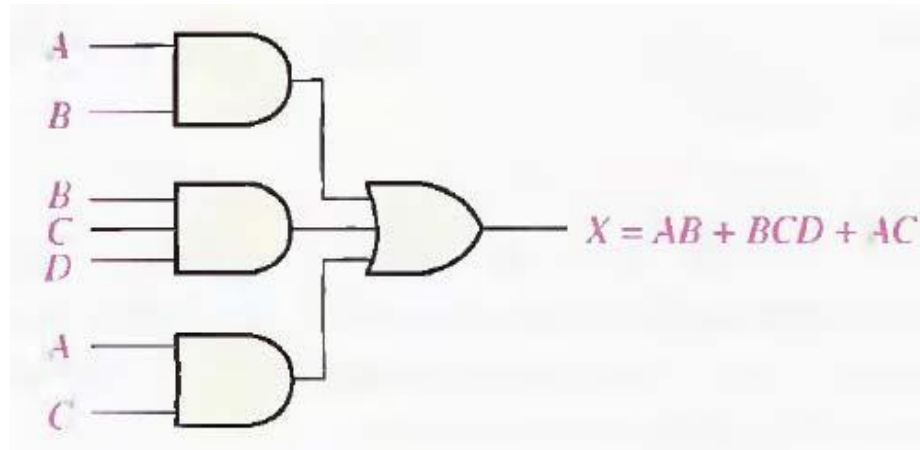
✓ $A\bar{B} + \bar{A}BC + AC$

- Domain of SOP expression: is the set of variables contain in the expression either in complemented and uncomplemented form
- Example: the domain of expression $A\bar{B} + \bar{A}BC = A, B, C$
- Example: the domain of expression $ABC + C\bar{D}E + \bar{B}C\bar{D} = A, B, C, D \text{ and } E$
- **In SOP, a single overbar cannot extend more than one variable, example :**

$A\bar{B} + \overline{ABC}$ ← not SOP because

SOP (Sum of Product)

- Implementing an SOP expression simply requires ORing the outputs of AND gates.
- A product term (minterm) is produced by AND operation.



- A standard SOP form must contain all of the variables in the domain of the expression for each product term, example:

$$\bar{A}\bar{B}C + \bar{A}BC + ABC$$

- Standard SOP forms are essential in constructing the truth table.
- In the following SOP form,

$$A\bar{B}C + \bar{A}\bar{B} + AB\bar{C}D$$

- How many minterms are there?
 $\Rightarrow 3$
- Is it a standard SOP form?
 $\Rightarrow$ No
- How do we convert a Boolean expression into standard SOP form?

SOP (Sum of Product)

- To convert SOP to its standard form, we use Boolean expression:

$$A + \bar{A}$$

$$A(B + C) = AB + AC$$

- Convert the following Boolean expression into SOP standard form
 $f(A, B, C) = AB'C + A'B'$

$$\begin{aligned} f(A, B, C) &= AB'C + A'B'(C + C') \\ &= AB'C + A'B'C + A'B'C' \end{aligned}$$

Example 6: Convert the following Boolean expression into standard SOP form.

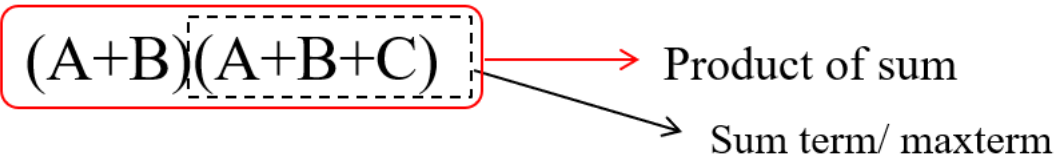
$$f(A, B, C, D) = AB'C + A'B' + ABC'D$$

Solution:

$$\begin{aligned} f(A, B, C) &= AB'CD + AB'CD' + A'B'CD \\ &\quad + A'B'CD' + A'B'C'D + A'B'C'D' \\ &\quad + ABC'D \end{aligned}$$

Standard (Canonical) Forms of Logic Expression

- POS (Product of Sum) is defined as two or more sum terms are multiplied. Example :

✓ $(A+B)(A+B+C)$ 

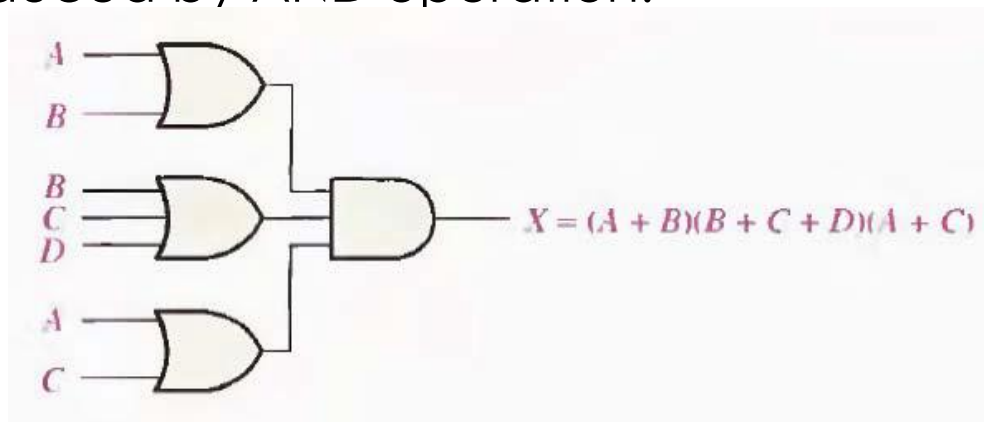
✓ $(A + \overline{B})(\overline{A} + B + \overline{C})(A + C)$

- In POS, a single over bar cannot extend more than one variable, example

$(A + \overline{B})(A + \overline{B + C})$  Not POS because $\overline{B + C}$

POS (Product of Sum)

- Implementing an POS expression simply requires ANDing the outputs of OR gates.
- A sum term (maxterm) is produced by OR operation. The product of sum terms is produced by AND operation.



- A standard POS form must contain all of the variables in the domain of the expression for each sum term, example:

$$(A + \bar{B} + C)(\bar{A} + B + C)(A + B + C)$$

- Standard POS forms are essential in constructing the truth table.
- In the following POS form,

$$(A + \bar{B} + C)(\bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

- How many maxterms are there?
 $\Rightarrow 3$
- Is it a standard POS form?
 $\Rightarrow$ No
- How do we convert a Boolean expression into standard POS form?

POS (Product of Sum)

- To convert POS to its standard form, we use Boolean expression:

$$A.\bar{A} = 0$$

$$A + BC = (A + B)(A + C)$$

- Convert the following Boolean expression into POS standard form
 $f(A, B, C) = (A + B' + C)(B' + C)$

$$\begin{aligned} f(A, B, C) &= (A + B' + C)(B' + C + A.A') \\ &= (A + B' + C)(A + B' + C)(A' + B' + C) \\ &= (A + B' + C)(A' + B' + C) \end{aligned}$$

Example 7: Convert the following Boolean expression into standard POS form.

$$\begin{aligned} f(A, B, C, D) \\ &= (A + B' + C)(B' + C + D')(A + B' + C' + D) \end{aligned}$$

Solution:

$$\begin{aligned} f(A, B, C, D) \\ &= (A + B' + C + D)(A + B' + C + D')(A' + B' + C + D')(A + B' + C' + D) \end{aligned}$$

SOP & POS - Exercise

- Convert the following Boolean expression into standard SOP form.

$$A(B + CD)$$

- Convert the following Boolean expression into standard POS form.

$$(A + \overline{B})(B + C)$$

Truth table based SOP and POS

- Develop a truth table for the SOP expression :

$$f(A, B, C) = A'B'C + AB'C' + ABC$$
- Domain = A, B, C , combinations = $2^3 = 8$
- Determine the inputs that makes the product term/minterm = 1.
- An SOP expression is equal to 1 only if one or more of the product terms in the expression is equal to 1
- Currently , 3 product terms :
 - $A'B'C$
 - $AB'C'$
 - ABC
- Finally, fills up the truth table

Input			Output	Product term
A	B	C	F	
0	0	0	0	
0	0	1	1	$\bar{A}\bar{B}C$
0	1	0	0	
0	1	1	0	
1	0	0	1	$A\bar{B}\bar{C}$
1	0	1	0	
1	1	0	0	
1	1	1	1	ABC

Truth table based SOP and POS

- Develop a truth table for the POS expression :
 $(A + B + C)(A + \overline{B} + C)(\overline{A} + B + \overline{C})$
- Domain = A, B, C : combinations = $2^3 = 8$
- Determine the inputs that makes the sum term/ maxterm = 0.
- An POS expression is equal to 0 only if one or more of the product terms in the expression is equal to 0
- Currently , 3 product terms :
 - $(A + B + C)$
 - $(A + B' + C)$
 - $(A' + B + C')$
- Finally, fills up the truth table

A	B	C	F	SUM TERM/ MAXTERM
0	0	0	0	$A+B+C$
0	0	1	1	
0	1	0	0	$A+\overline{B}+C$
0	1	1	1	
1	0	0	1	
1	0	1	0	$\overline{A}+B+\overline{C}$
1	1	0	1	
1	1	1	1	

Truth table based SOP and POS

Example 8: Make a truth table for the following functions:

$$a) F = AB + A\overline{C}$$

$$b) F = (A + C)(B + \overline{C})$$

Truth table to SOP and POS

- Standard SOP expression simply from truth table.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

← A'BC

← AB'C'

← AB'C

← ABC'

← ABC

Note that each term has ALL variables. If a product term has ALL variables present, it is a **MINTERM**.

$$F(A,B,C) = A'BC + AB'C' + AB'C + ABC' + ABC$$

Truth table to SOP and POS

- Standard POS expression simply from truth table.

A	B	C	F
0	0	0	0 ← $A+B+C$
0	0	1	0 ← $A+B+C'$
0	1	0	0 ← $A+B'+C$
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Note that each term has ALL variables. If a sum term has ALL variables present, it is a **MAXTERM**.

$$F(A,B,C) = (A+B+C)(A+B+C')(A+B'+C)$$

Truth table –Minterm & Maxterm notation

- Each line in a truth table represents both a Minterm and Maxterm

Row	Input			SOP Form	POS Form
	A	B	C	Minterm	Maxterm
0	0	0	0	$A'B'C' = m_0$	$A+B+C = M_0$
1	0	0	1	$A'B'C = m_1$	$A+B+C' = M_1$
2	0	1	0	$A'BC' = m_2$	$A+B'+C = M_2$
3	0	1	1	$A'BC = m_3$	$A+B'+C' = M_3$
4	1	0	0	$AB'C' = m_4$	$A'+B+C = M_4$
5	1	0	1	$AB'C = m_5$	$A'+B+C' = M_5$
6	1	1	0	$ABC' = m_6$	$A'+B'+C = M_6$
7	1	1	1	$ABC = m_7$	$A'+B'+C' = M_7$

Truth table –Minterm & Maxterm notation

- A Boolean function can be written in terms of Minterm or Maxterm notation as shorthand method of specifying the function.

Input			Output	minterm/ maxterm
A	B	C	F	
0	0	0	0	$A + B + C$
0	0	1	0	$A + B + \bar{C}$
0	1	0	0	$A + \bar{B} + C$
0	1	1	1	$\bar{A}BC$
1	0	0	1	$A\bar{B}\bar{C}$
1	0	1	1	$A\bar{B}C$
1	1	0	1	$AB\bar{C}$
1	1	1	1	ABC

* **Minterms** correspond to '1' of F, **Maxterms** correspond to '0' of F in truth table.

Minterm notation:

$$\begin{aligned}
 F(A, B, C) &= \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC \\
 &= m_3 + m_4 + m_5 + m_6 + m_7 \\
 &= \sum m(3, 4, 5, 6, 7)
 \end{aligned}$$

Maxterm notation:

$$\begin{aligned}
 F(A, B, C) &= (A + B + C)(A + B + \bar{C})(A + \bar{B} + C) \\
 &= M_0 \cdot M_1 \cdot M_2 \\
 &= \prod M(0, 1, 2)
 \end{aligned}$$

Truth table –Minterm & Maxterm notation

Example 9: Given minterm $F(A, B, C) = \sum m(1, 2, 6)$ and maxterm $F(A, B, C) = \sum M(0, 3, 4, 5, 7)$ forms. Find the truth table.

Solution:

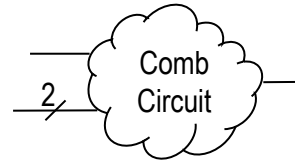
Minterm/ Maxterm notation	Input			Output
	A	B	C	F
M_0	0	0	0	0
m_1	0	0	1	1
m_2	0	1	0	1
M_3	0	1	1	0
M_4	1	0	0	0
M_5	1	0	1	0
m_6	1	1	0	1
M_7	1	1	1	0

Digital Circuit Structure Classification

- Generally, digital circuits are classified into:

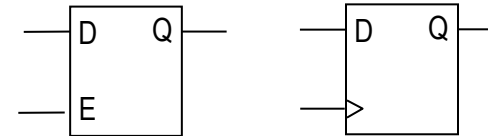
1. Combinational circuit

- ✓ Asynchronous (non-clocked).



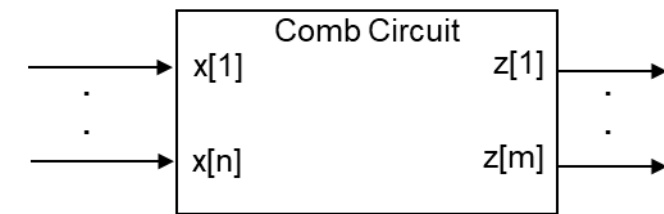
2. Memory element circuit.

- ✓ Asynchronous (non-clocked) – latch circuits.
- ✓ Synchronous (clocked) – flip-flop circuits.

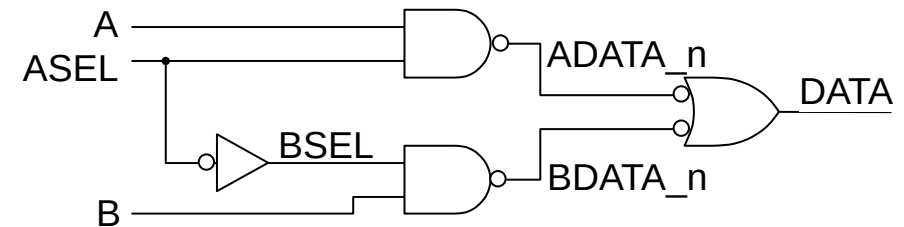


Combinational circuit

- **Definition** - Combinational circuit - the output is a function of only the present input.
- The timing behaviour of a combinational circuit can be described/ represented/ modelled using:
 - i. Boolean algebra.
 - $z[m] = f_m(x[1], \dots, x[n])$, where $m = 1$ to k
 - ii. Truth table.
 - iii. Timing diagram.
 - iv. Block diagram and schematic diagram.
 - v. HDL – Hardware Description Language
- The timing behaviour can be modelled using the timing diagram.



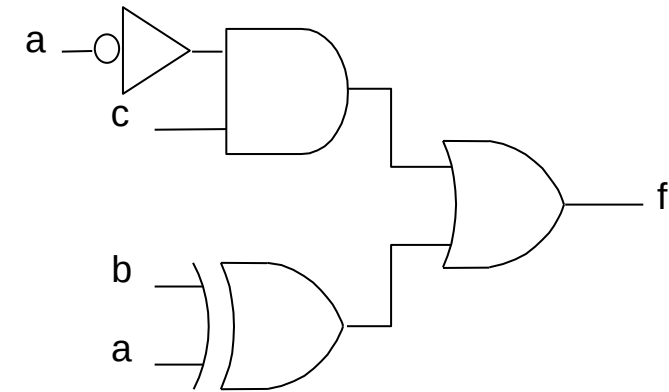
Block diagram of a combinational circuit.



Combinational circuit analysis

- Combination circuit analysis enable us to determine the functions that the circuits implement.
- Analysis is useful to:
 - Determine the behavior of the circuit
 - ✓ Logic equations, truth tables, timing diagrams and schematic are used to describe the behavior of the logic circuit.
 - Verify the correctness of the circuit (as compared to the circuit specification).
 - Assist in converting the circuit to a different form.

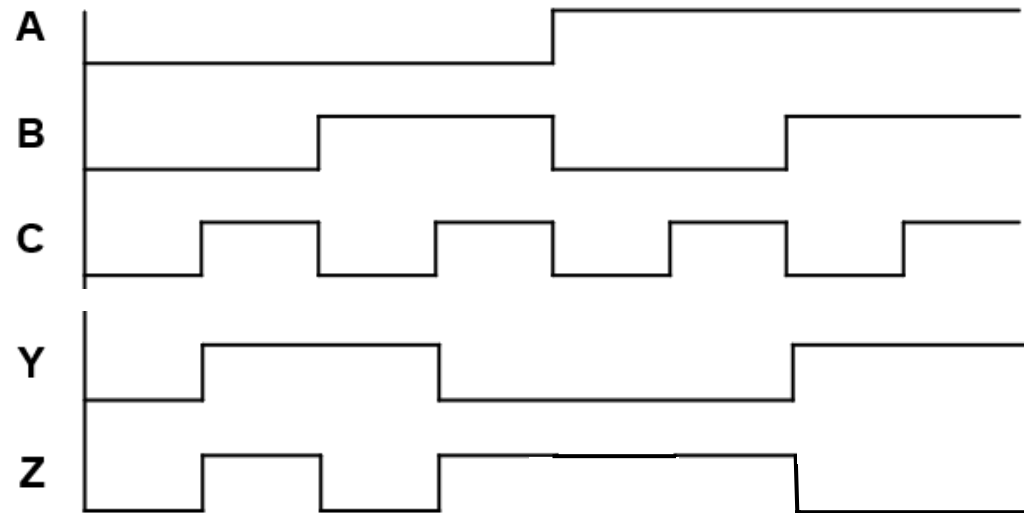
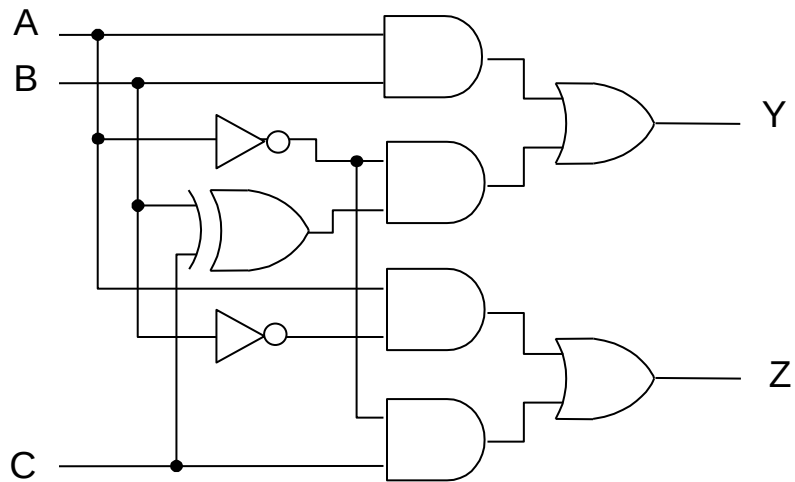
Exercise: Describe the behaviour of the logic circuit in the form of truth table.



Combinational circuit analysis

Exercise:

- Sketch the output Y and Z based on the circuit given below
- Find the truth table and the minterm lists for Y and Z.



Combinational circuit analysis

Solution:

Inputs	Outputs	
A B C	Y	Z
0 0 0	0	0
0 0 1	1	1
0 1 0	1	0
0 1 1	0	1
1 0 0	0	1
1 0 1	0	1
1 1 0	1	0
1 1 1	1	0

$$Y = A'B'C + A'BC' + ABC' + ABC$$

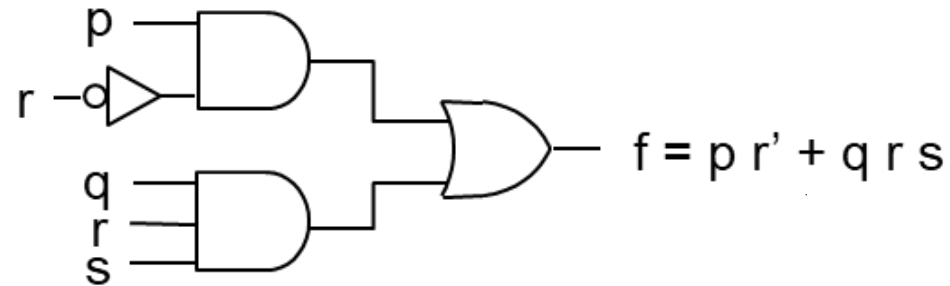
$$= \sum m(1, 2, 6, 7)$$

$$Z = A'B'C + A'BC + AB'C' + AB'C$$

$$= \sum m(1, 3, 4, 5)$$

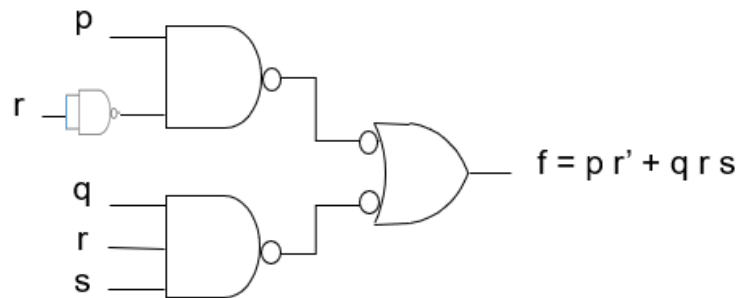
Combinational circuit analysis

Exercise: Convert the SOP circuit (AOI) to all NAND gates circuit



Solution:

Method 1:



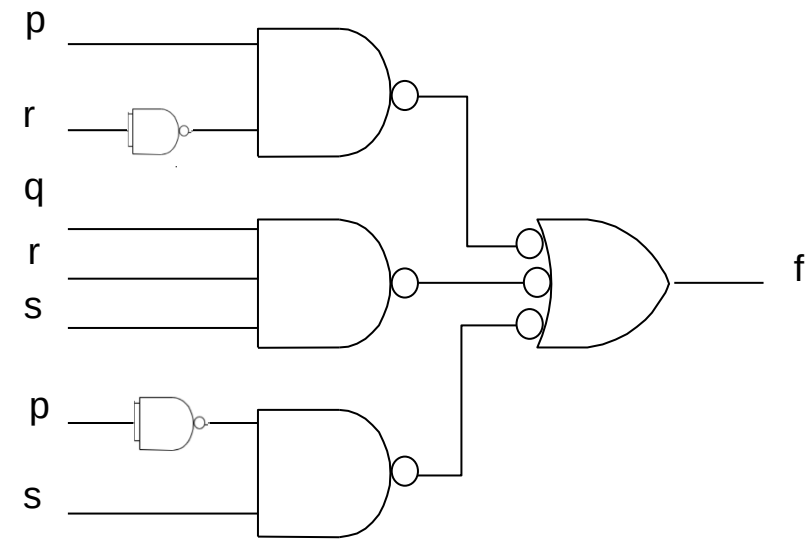
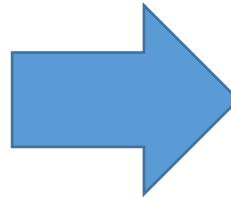
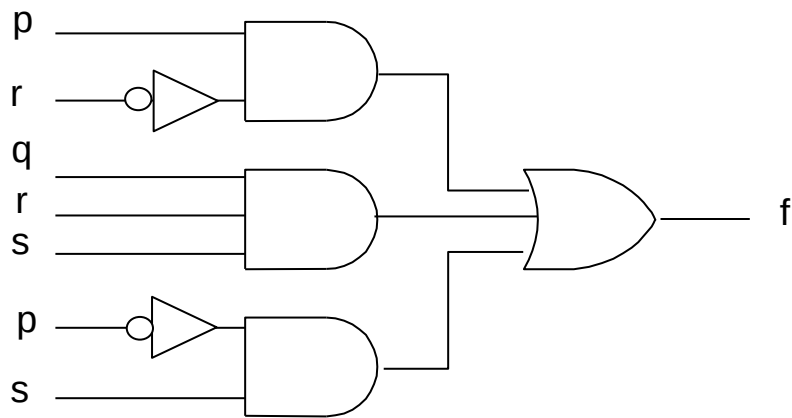
Method 2:

$$\begin{aligned}
 f &= pr' + qrs \\
 &= (pr' + qrs)'' \quad (\text{DeMorgan's theorem}) \\
 &= ((pr')'(qrs)')' \quad (\text{NAND form})
 \end{aligned}$$

After we have obtained the NAND form, draw the circuit

Combinational circuit analysis

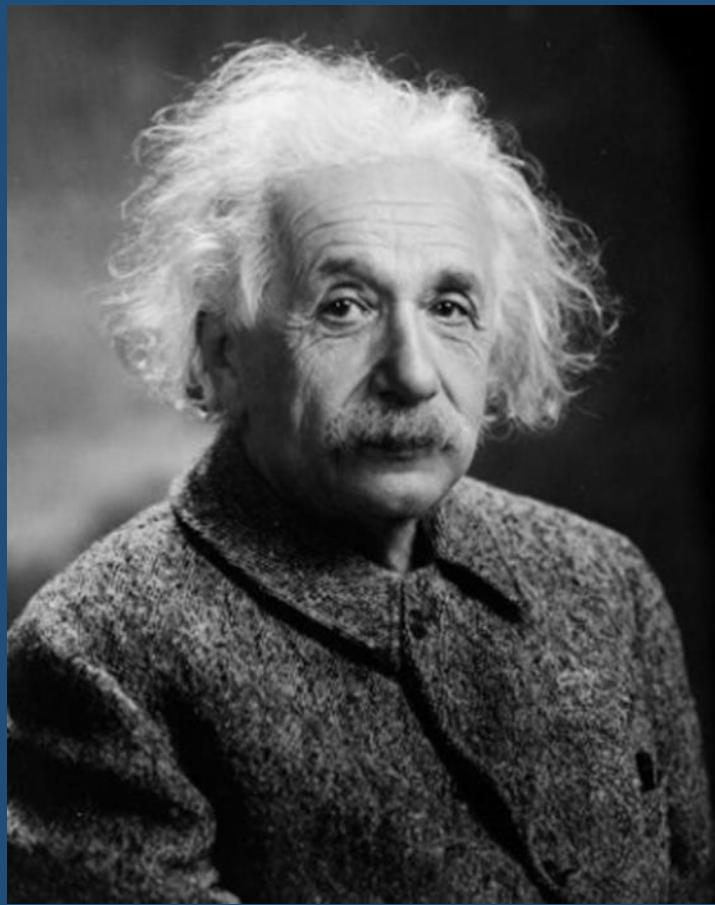
Exercise: Convert the SOP circuit (AOI) to all NAND gates circuit



- For reading, analysing and modelling work, digital circuits are expressed in terms of AND and OR gates.
 - For actual implementation on silicon (to prepare for physical design), use NAND and NOR.
- ✓ **Faster than AND and OR**

Combinational circuit analysis

Exercise: The input to a logic circuit consists of four signals A, B, C and D. These inputs represent 4-bit binary number where A is the MSB and D is the LSB. Use algebraic method to design a logic circuit such that the output is high only when the binary number input is less than 7. Design the circuit using only NAND gates.



“Creativity is
intelligence
having fun”

– Albert Einstein



Q & A

Thank you